TEK-UP
Ecole Supérieure Privée
Technologie & Ingénierie

AWS ARCHITECTURE

Elastic Load Balancing & Auto Scaling Groups



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Course Plan



Elastic Load Balancing ELB

Auto Scaling Groups

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Scalability & High Availability

- Scalability means that an application / system can handle greater loads by adapting.

- There are two kinds of scalability :



- **Scalability is linked but different to High Availability**

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Vertical Scalability

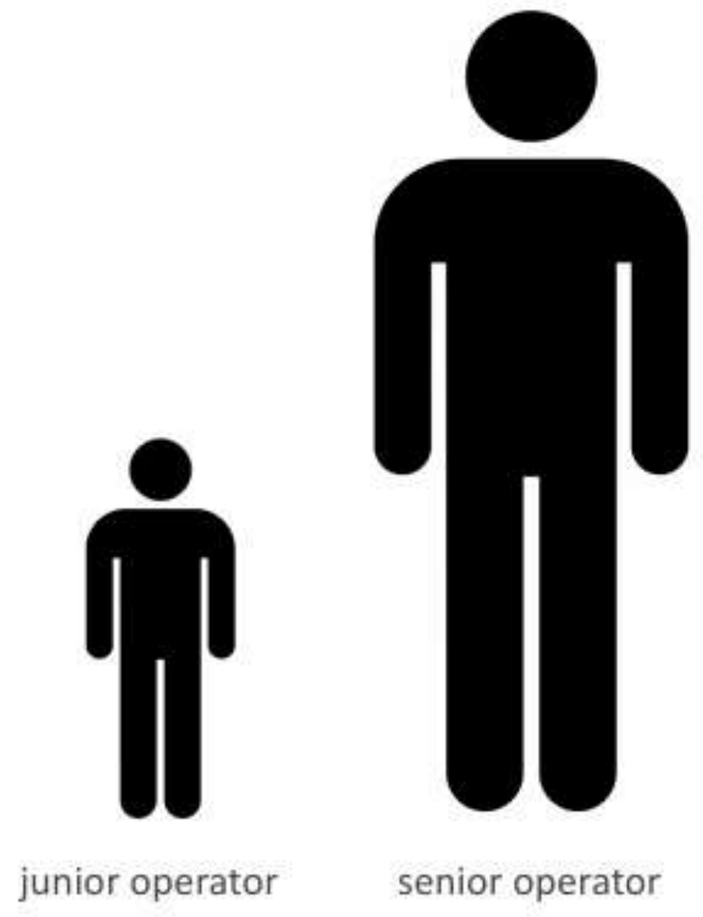
Vertical Scalability means increasing the size of the instance

Example :

- your application runs on a t2.micro
- Scaling that application vertically means running it on a t2.large
- Vertical scalability is very common for non distributed systems, such as a database.

➢ There is usually a limit to how much you can vertically scale

→ **hardware limit**



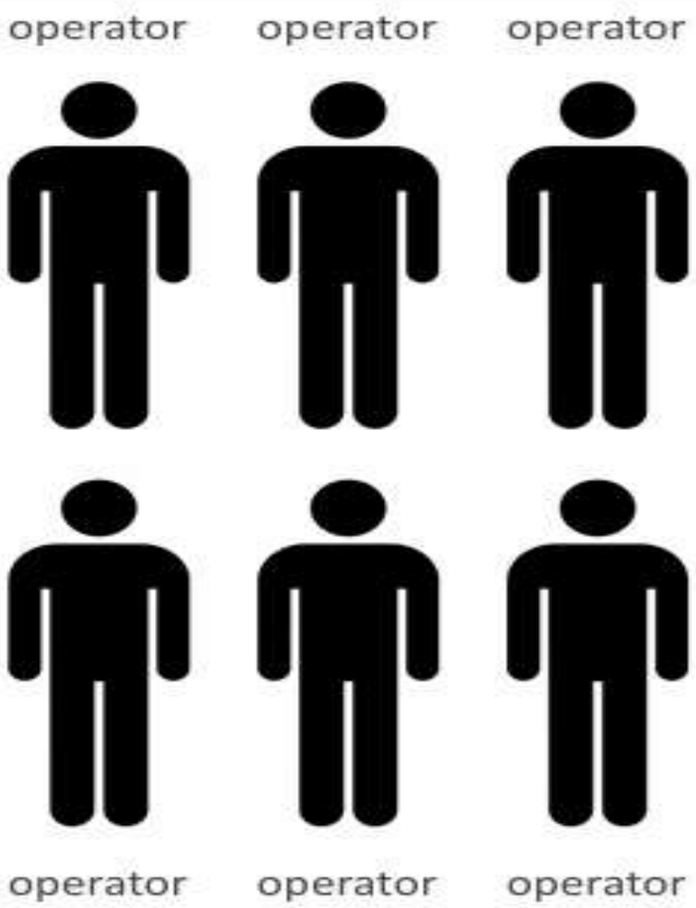
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Horizontal Scalability

Horizontal Scalability means increasing the number of instances / systems for your application.

- Horizontal scaling implies distributed systems
- This is very common for web applications / modern applications

➢ It's easy to horizontally scale thanks the cloud offerings such as Amazon EC2



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High Availability

- **High Availability** usually goes hand in hand with horizontal scaling.

- High availability means running your application / system in **at least** 2 Availability Zones

- The goal of high availability is **to survive** a data center loss (disaster





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High Availability & Scalability For EC2

- **Vertical Scaling :** Increase instance size (= **scale up / down**).
  - **From** : t2.nano - 0.5G of RAM, 1 vCPU
  - **To** : u-12tb1.metal – 12.3 TB of RAM, 448 vCPUs

- **Horizontal Scaling :** Increase number of instances (= **scale out** / **scale in**)
  - Auto Scaling Group
  - Load Balancer

- **High Availability :** Run instances for the same application across multi AZ
  - Auto Scaling Group multi AZ
  - Load Balancer multi AZ

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Scalability vs Elasticity (vs Agility)

- **Scalability :** ability to accommodate a larger load by making the hardware stronger (scale up), or by adding nodes (scale out)

- **Elasticity :** once a system is scalable, elasticity means that there will be some “auto-scaling” so that the system can scale based on the load. This is “cloud-friendly”: pay-per-use, match demand, optimize costs

- **Agility :** (not related to scalability) new IT resources are only a click away, which means that you reduce the time to make those resources available to your developers from weeks to just minutes.

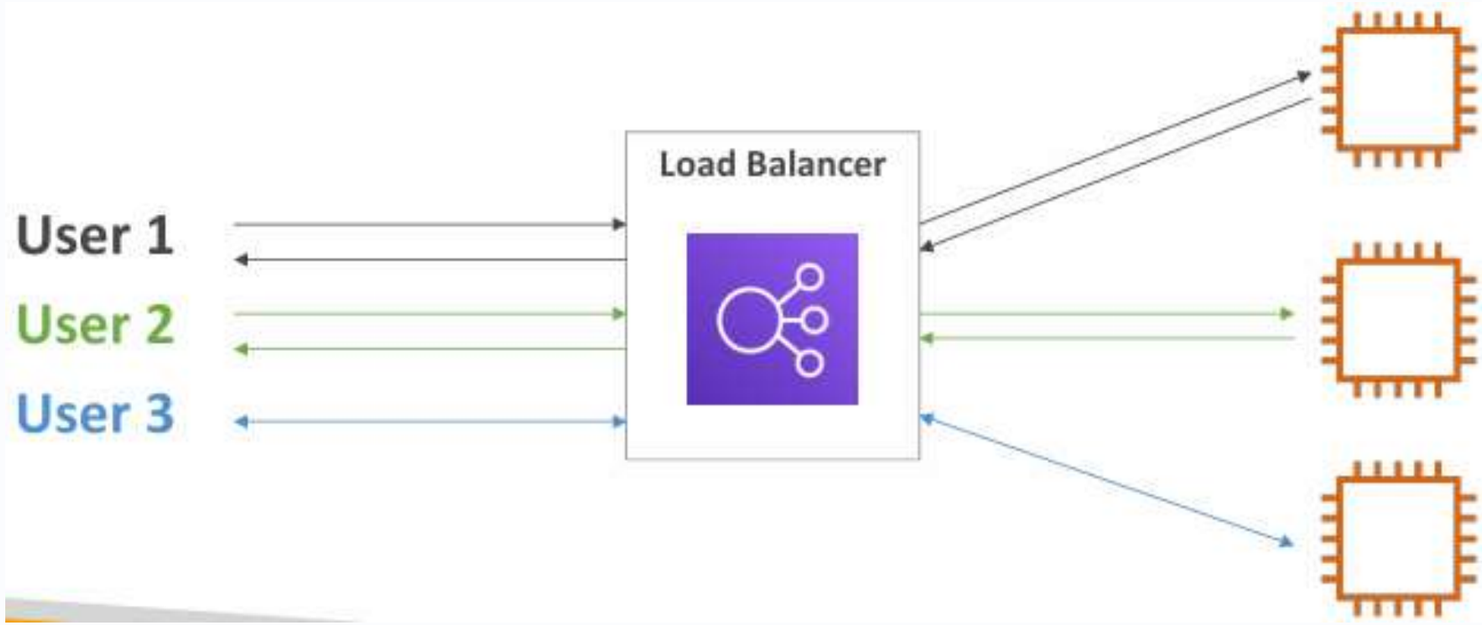
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What is load balancing ?





➢ Load balancers are servers that forward internet traffic to multiple servers (EC2 Instances) downstream.



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Why use a load balancer ?



- Spread load across multiple downstream instances
- Expose a single point of access (DNS) to your application
- Seamlessly handle failures of downstream instances
- Do regular health checks to your instances
- Provide SSL termination (HTTPS) for your websites
- High availability across zones

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Why use a Elastic Load balancer ?

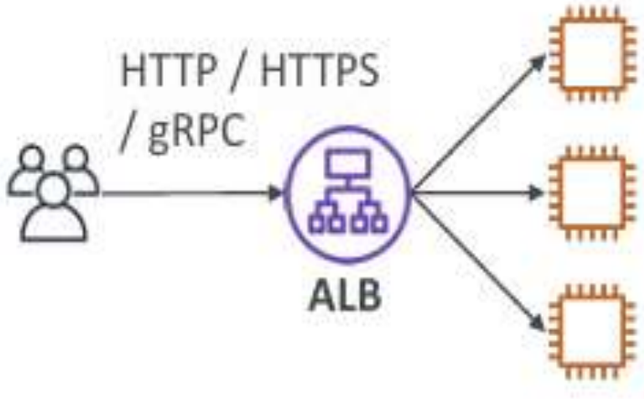


- An **ELB** (Elastic Load Balancer) is a **managed load balancer**
  - AWS guarantees that it will be working
  - AWS takes care of upgrades, maintenance, high availability
  - AWS provides only a few configuration knobs
- It costs less to setup your own load balancer but it will be a lot more effort on your end (maintenance, integrations)
- 4 kinds of load balancers offered by AWS :
  - **Application Load Balancer** (HTTP / HTTPS only) – Layer 7
  - **Network Load Balancer** (ultra-high performance, allows for TCP) – Layer 4
  - **Gateway Load Balancer** – Layer 3
  - **Classic Load Balancer** (retired in 2023) – Layer 4 & 7

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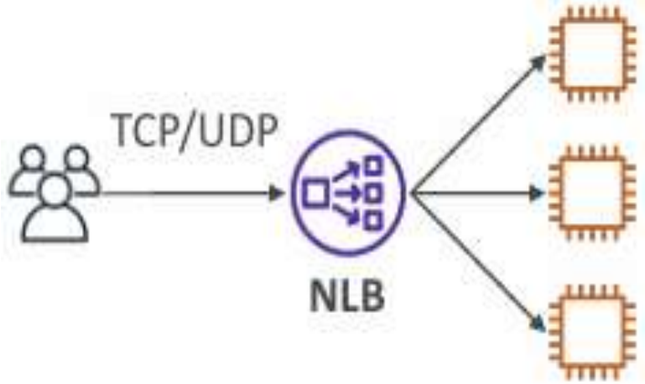
Application Load Balancer

- HTTP/HTTPS/gRPC (Layer 7)
- **HTTP Routing** features
- **Static DNS** (URL)



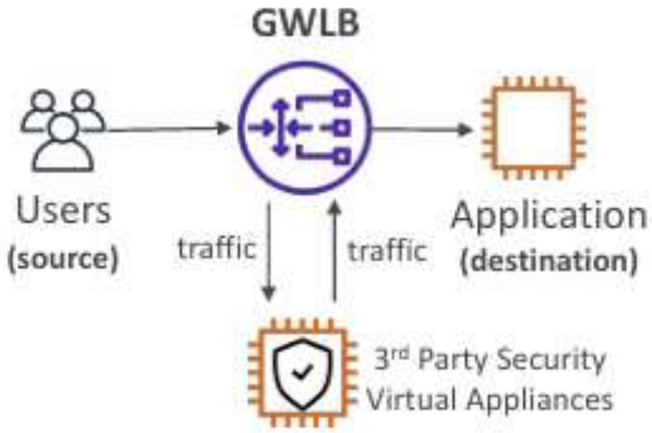
Network Load Balancer

- TCP/UDP (Layer 4)
- High Performance: millions of request per seconds
- **Static IP** through Elastic IP



Gateway Load Balancer

- GENEVE on IP Packets(Layer 3)
- Route Traffic to Firewalls that you manage on EC2 instances



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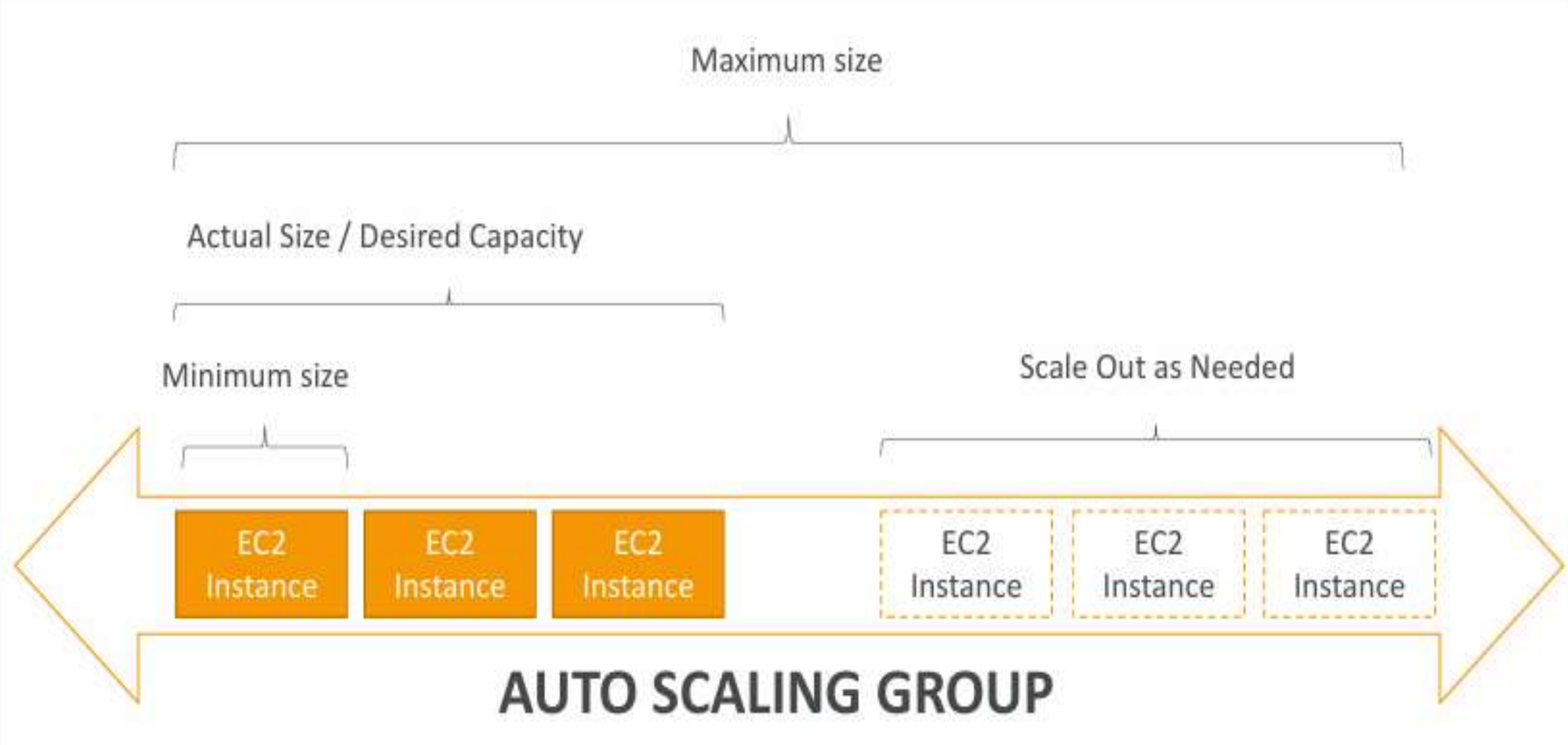
What’s an Auto Scaling Group ?



- In real-life, the load on your websites and application can change
- In the cloud, you can create and get rid of servers very quickly
- The goal of an Auto Scaling Group (ASG) is to :
    - Scale out (add EC2 instances) to match an increased load
    - Scale in (remove EC2 instances) to match a decreased load
    - Ensure we have a minimum and a maximum number of machines running
    - Automatically register new instances to a load balancer
    - Replace unhealthy instances
- Cost Savings: only run at an optimal capacity (principle of the cloud)

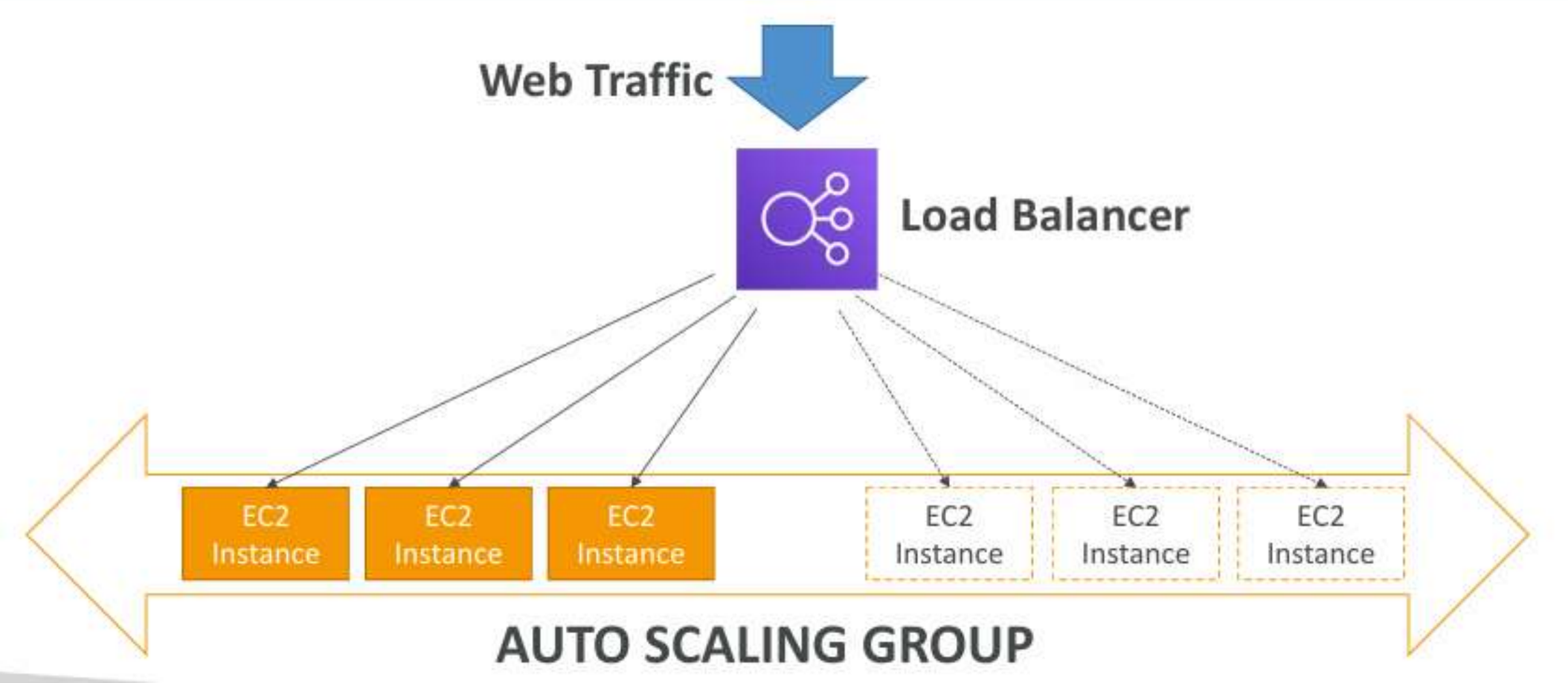
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Auto Scaling Group in AWS



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ASG in AWS With Load Balancer



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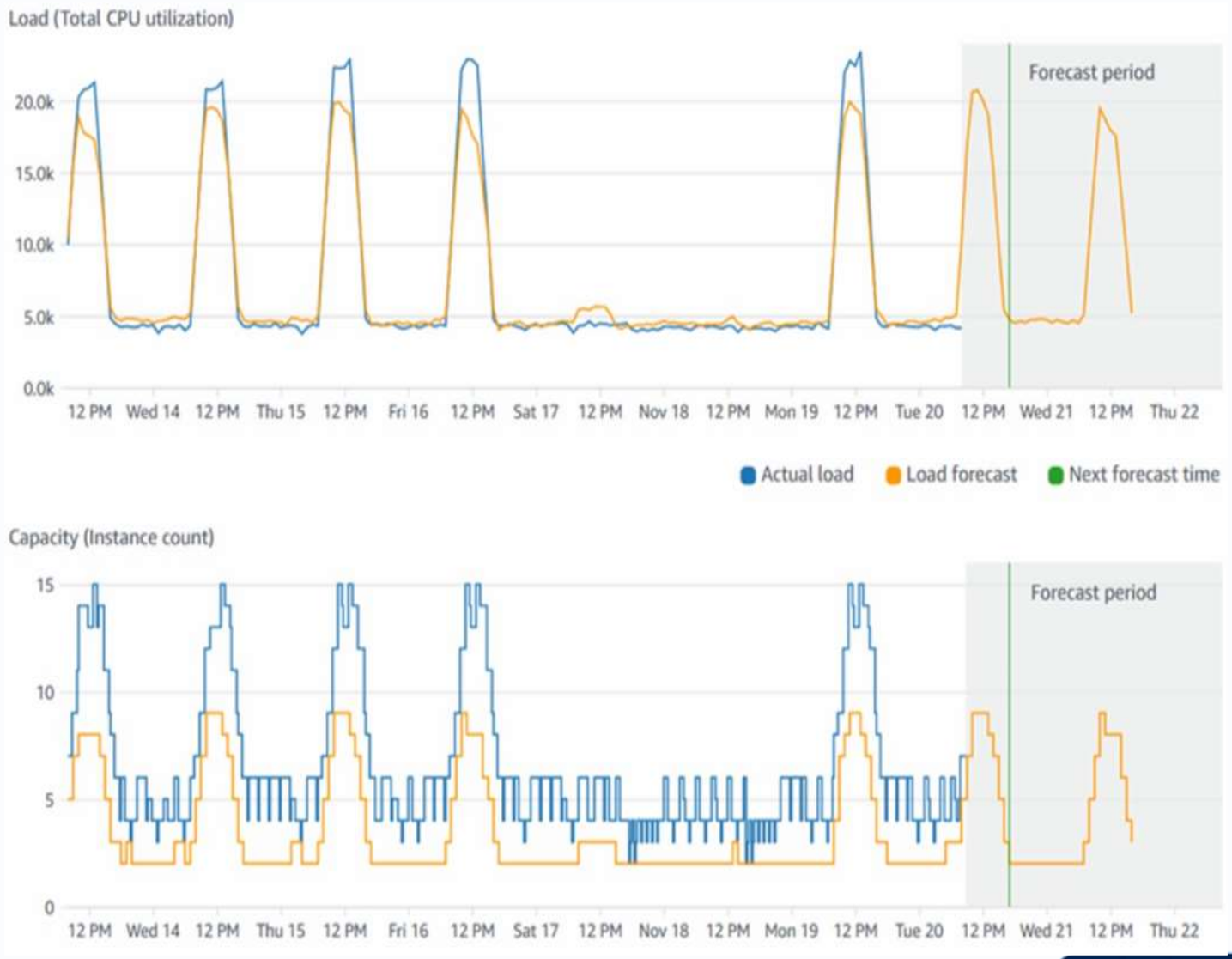
Auto Scaling Groups – Scaling Strategies

- **Manual Scaling :** Update the size of an ASG manually
- **Dynamic Scaling :** Respond to changing demand
  - **Simple / Step Scaling**
    - When a CloudWatch alarm is triggered (example CPU > 70%), then add 2 units
    - When a CloudWatch alarm is triggered (example CPU < 30%), then remove 1
  - **Target Tracking Scaling**
    - Example: I want the average ASG CPU to stay at around 40%
  - **Scheduled Scaling**
    - Anticipate a scaling based on known usage patterns
    - Example: increase the min. capacity to 10 at 5 pm on Fridays

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Auto Scaling Groups – Scaling Strategies

- **Predictive Scaling :**
  - Uses Machine Learning to predict future traffic ahead of time
  - Automatically provisions the right number of EC2 instances in Advance



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ELB & ASG – Summary

- **High Availability** vs **Scalability** (vertical and horizontal) vs **Elasticity** vs **Agility** in the Cloud
- **Elastic Load Balancers (ELB) :**
  - Distribute traffic across backend EC2 instances, can be Multi-AZ
  - Supports health checks
  - 4 types: Classic (old), Application (HTTP – L7), Network (TCP – L4), Gateway (L3)
- **Auto Scaling Groups (ASG) :**
  - Implement Elasticity for your application, across multiple AZ
  - Scale EC2 instances based on the demand on your system, replace unhealthy
  - Integrated with the ELB

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